TA212 LAB

PROJECT REPORT, 1ST SEMESTER 2025-26

MECHANICAL INTEGRATOR

Group No. – 9

Lab Day – TUESDAY

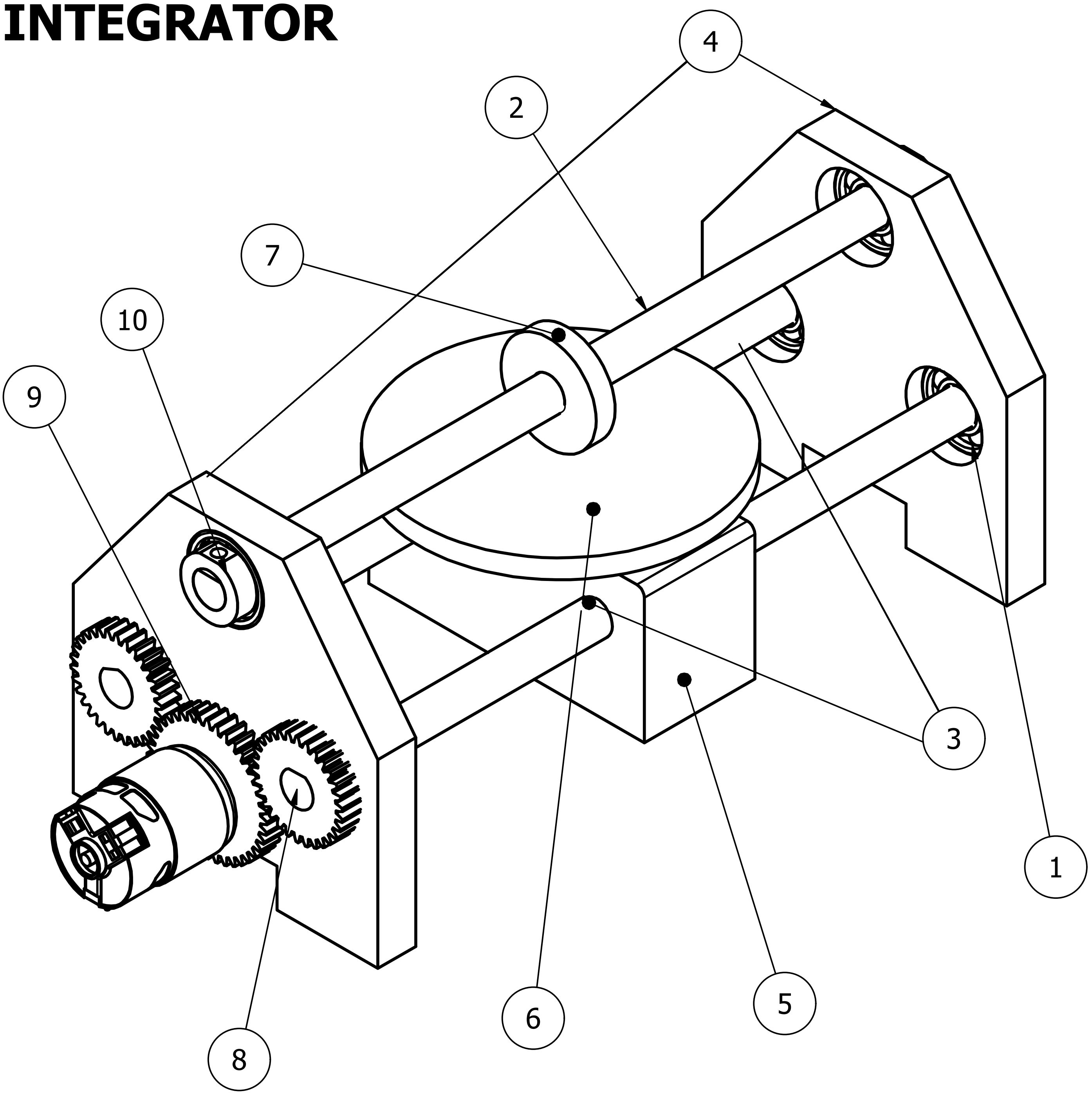
Course Instructor – Dr. Sarvesh Mishra

Lab Guide – Arun Kumar Dubey

Group Members

Sl. No.	Name	Roll No.
1.	Adish Sudhir Kapate	240046
2.	B Mukund Advaith	240253
3.	Daksh M Jain	240319
4.	Pratik Beniwal	240783
5.	Priyanshu Mehta	240808
6.	Rohith Rajeev Nambiar	240882
7.	S Kiran	240895
8.	Vegesna Ram Charan Varma	241156

MECHANICAL INTEGRATOR



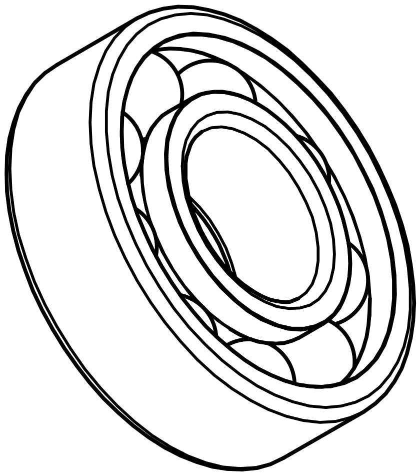
SCALE 1:2.2

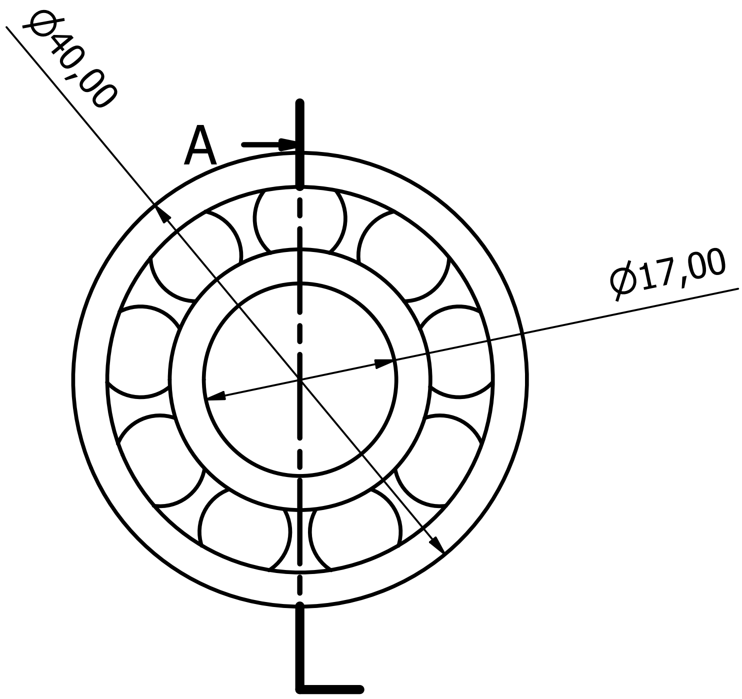
DRAWN Kiran	02-09-2025	
CHECKED		TITLE
QA		
MFG		
APPROVED		
		SIZE A4 / DWG NO Project_DWG / REV
		SCALE 1 / 2.2 / SHEET 1 OF 1

INDEX / PART LIST

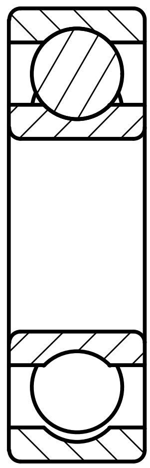
Sl. No.	Part No.	Part Name	Qty	Dimensions in mm	Material	Pg. No.
1	-	Cover Page	-	-	-	1
2	-	Complete Assembly	-	-	-	2
3	-	Index / Part List	-	-	-	3
4	1	Ball Bearing	6	dia 40 x 12	-	4
5	2	Stationary Rod	1	dia 17 x 370	MS	5
6	3	Drive Shaft	2	dia 17.2 x 370	MS	6
7	4	Support Stand	2	160 x 161 x 20	MS	7
8	5	Carriage / Mover	1	145 x 58 x 70	MS	8
9	6	Large Base Disk	1	dia 148 x 50	PLA	9
10	7	Stationary Wheel	1	dia 50 x 20	PLA	10
11	8	Spur Gear (n=30)	2	dia 48 x 20	MS	11
12	9	Spur Gear (n=40)	1	dia 63 x 20	MS	12
13	10	Stationary Rod Stopper	4	dia 30 x 10	MS	13

PART 1





A-A (1.5 : 1)



Dimensions in mm
BEARING NUMBER: 6203
QUANTITY: 6

Designed by	Checked by	Approved by	Date		Date	
Kiran					09-09-2025	

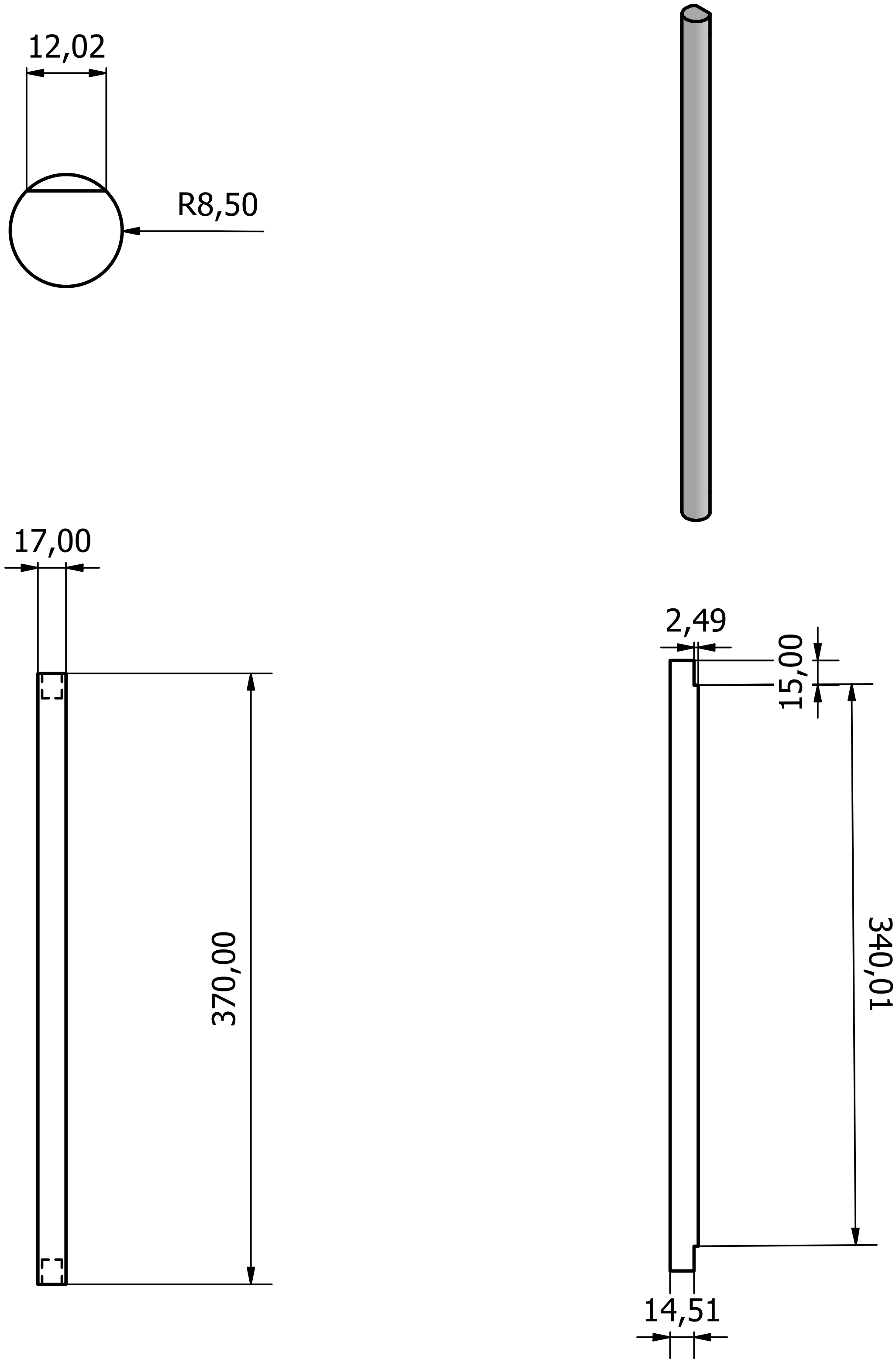
	Edition	Sheet
Ball Bearing 6203		1 / 1

PART - 2

PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	Project_stationary_rod

Scale = 3:1
Process:-
Turning/Lathing
Materials:-
Mild Steel

All Dimensions in mm unless specified otherwise

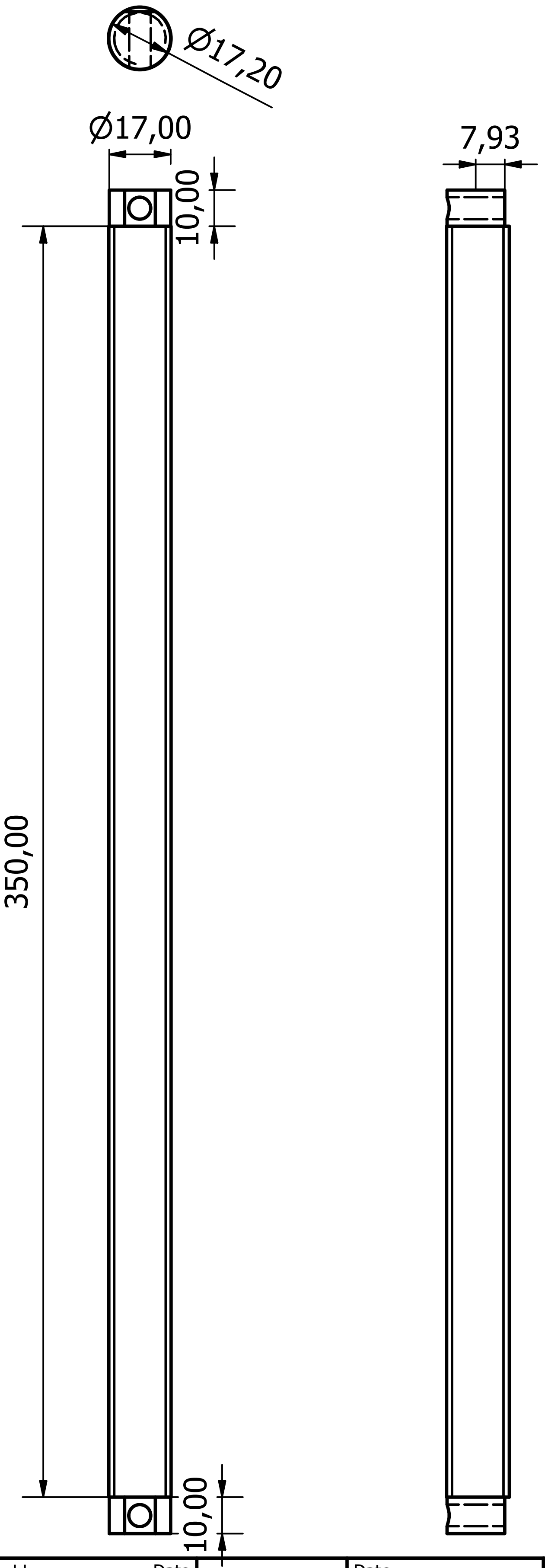


Designed by	Checked by	Approved by	Date		Date	
devku					09-09-2025	

Project_stationary_rod[1]	Edition	Sheet 1 / 1

PART - 3

All Dimensions in mm
PROCESS: Turning, Drilling, Milling
Threading:
SIZE: 8
CLASS: 6H
DEPTH: 1mm
ORIENTATION: R
LENGTH: 350mm
SCALE 1 : 1.8



QUANTITY: 2
MATERIAL: Mild Steel

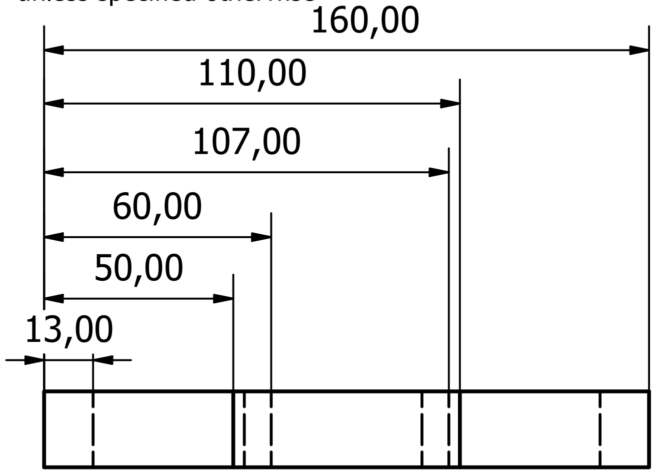
Designed by	Checked by	Approved by	Date		Date	
Kiran					09-09-2025	

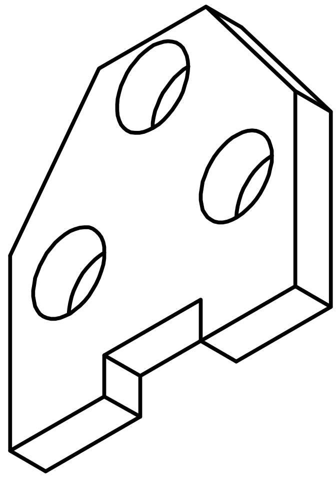
	Edition	Sheet
Proj_drive_shaft		1 / 1

PARTS LIST		
ITEM	QTY	PART NUMBER
1	2	4

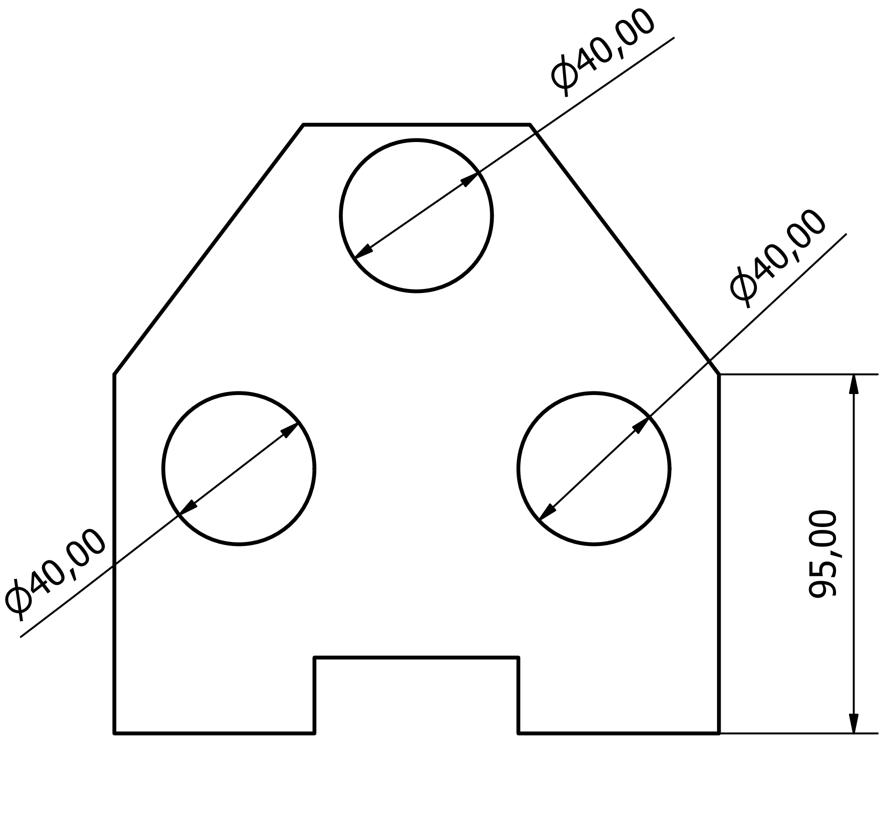
SCALE = 1:3
DIMENSION : 161 x 160
MATERIAL : M.S. SHEET(20MM)
PROCESS : CONVENTIONAL (DRILLING, MILLING, CUTTING, TURNING/LATHING)

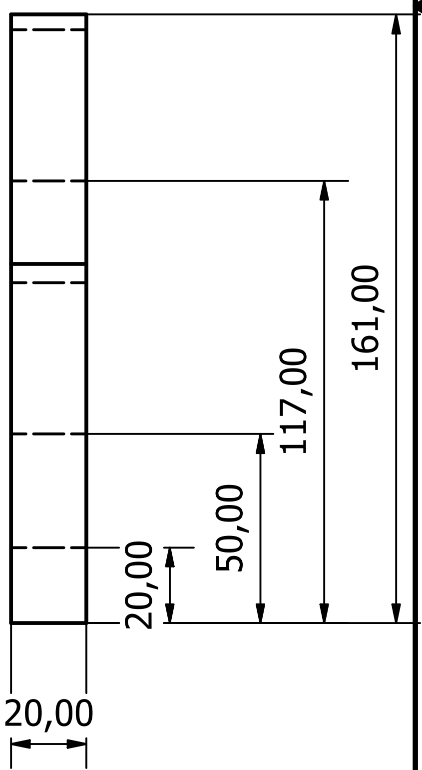
All dimensions given in mm,
unless specified otherwise





ISOMETRIC





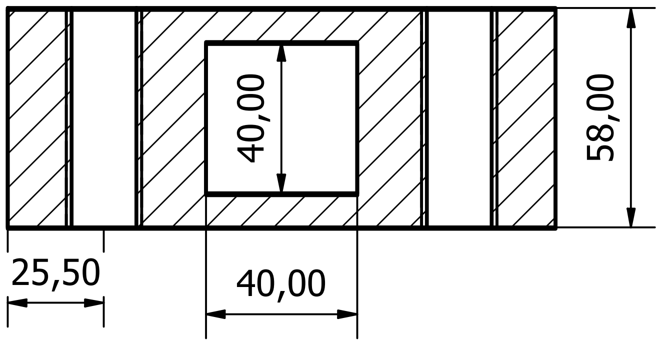
Designed by	Checked by	Approved by	Date	Date	
Asus				08-09-2025	

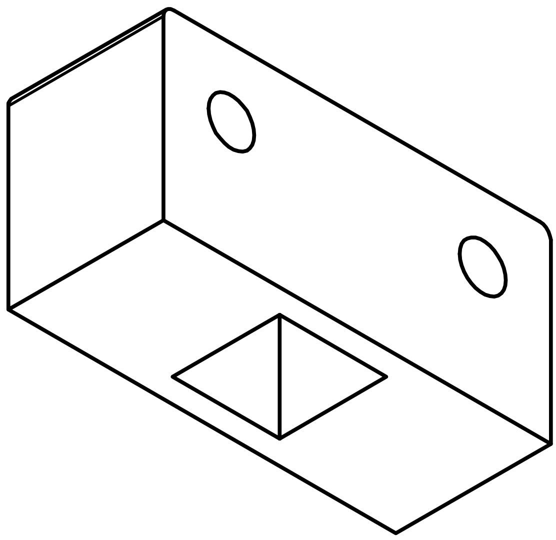
	Edition	Sheet
Project_support_stand_final		1 / 1

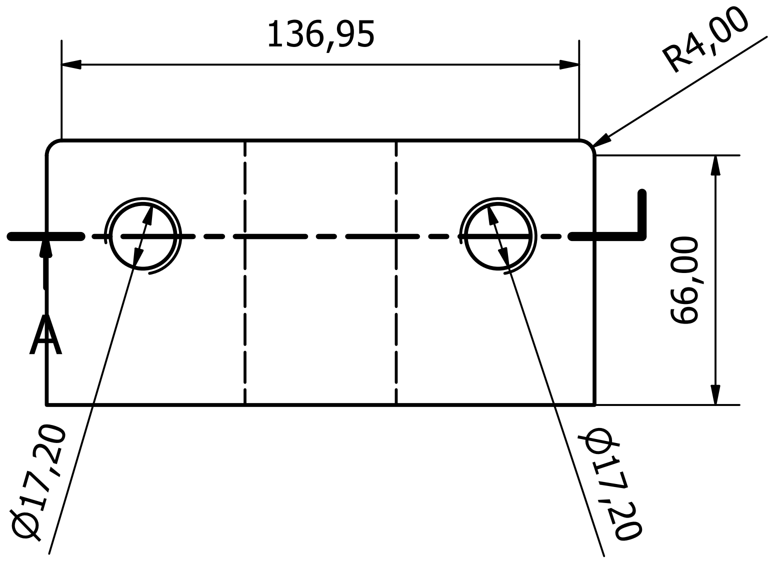
PART - 5

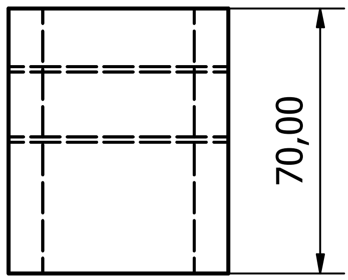
SCALE = 1:2
THREADING:
Class: 6H
Size: 18 (ISO Metric)

PROCESS: Drilling,
Milling, Cutting









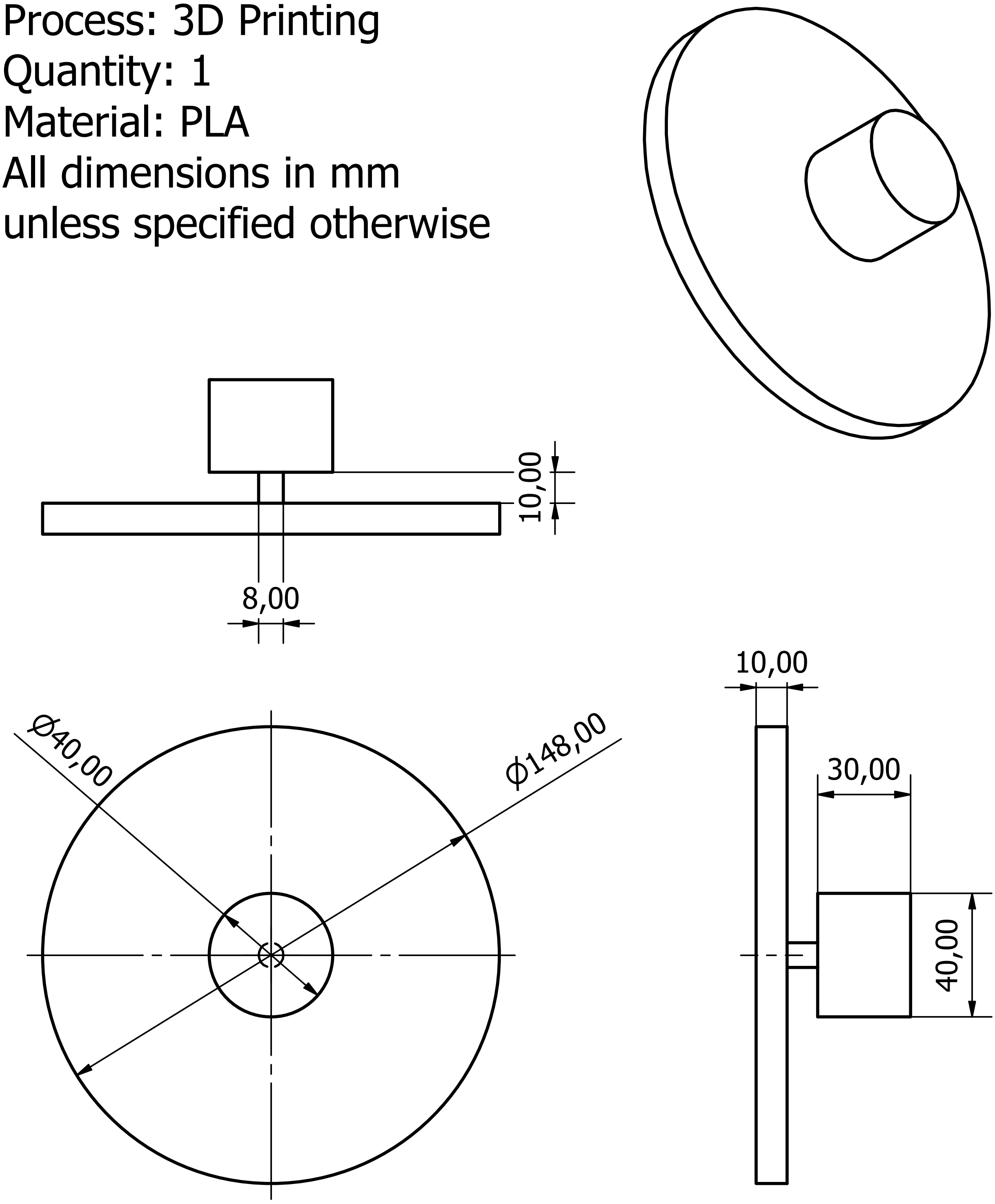
All dimensions in mm unless
specified otherwise

QUANTITY: 1
Material: MILD STEEL

DRAWN Kiran	03-09-2025				
CHECKED		TITLE			
QA					
MFG					
APPROVED					
		SIZE A4		DWG NO project_rough_mover	REV
		SCALE 1 : 2			SHEET 1 OF 1

PART - 6

SCALE = 1:2
Process: 3D Printing
Quantity: 1
Material: PLA
All dimensions in mm
unless specified otherwise



Designed by	Checked by	Approved by	Date		Date	
mehta					08-09-2025	

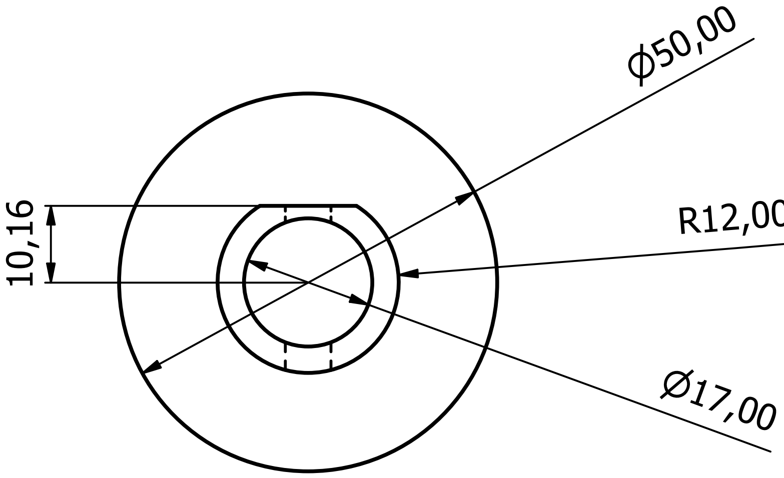
	Edition	Sheet
Project_Rough_disc[1]		1 / 1

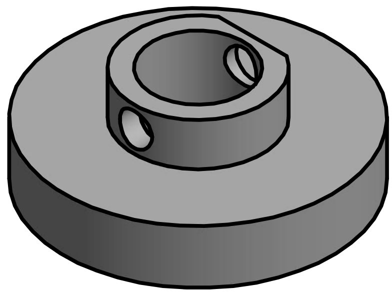
PART - 7

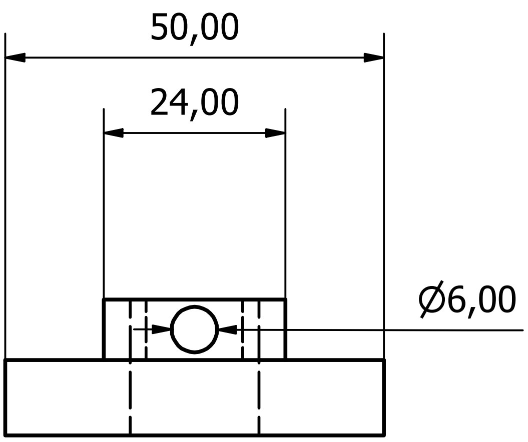
PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	Project_Stationary_Wheel[1]

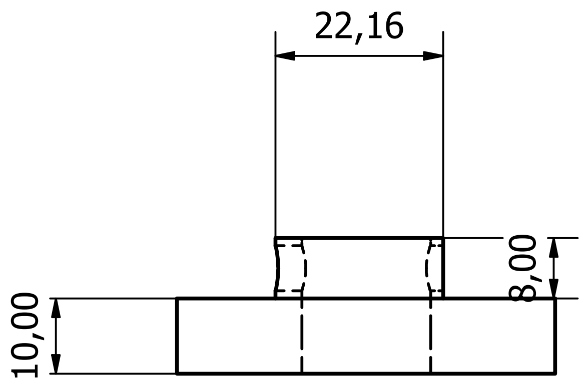
Scale 1:1
Process:-
3D Printing
Materials:-
PLA

All dimensions in mm unless specified otherwise





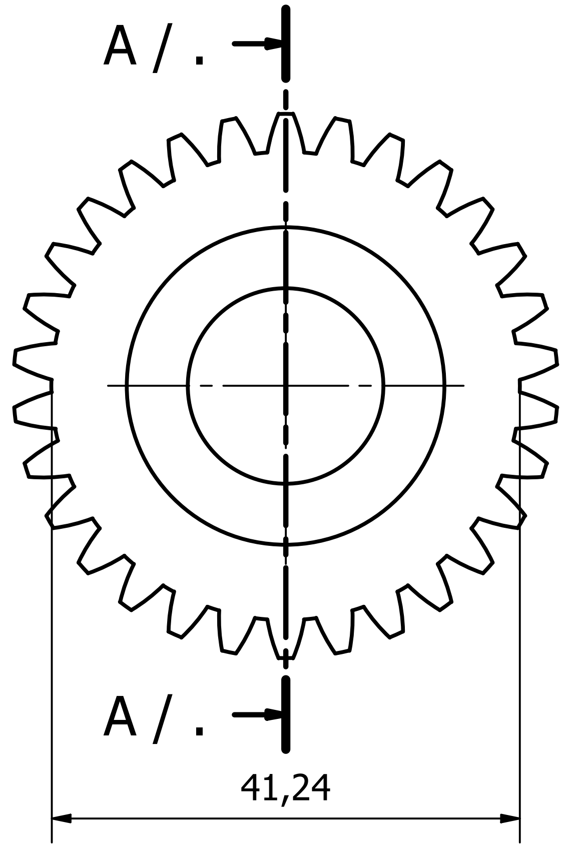




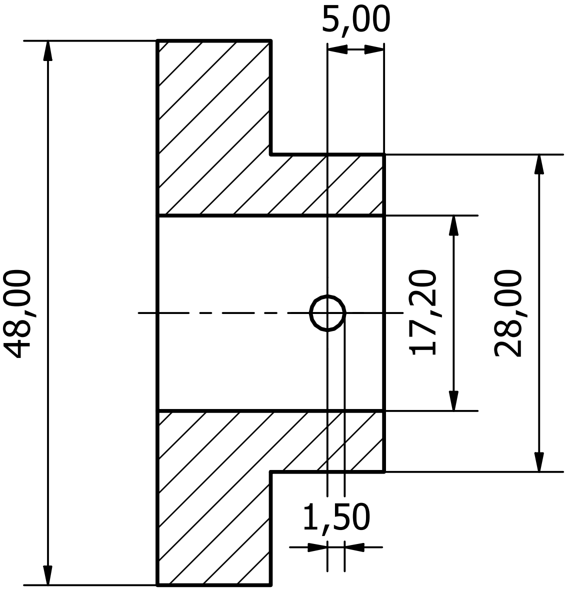
Designed by	Checked by	Approved by	Date	Date	
devku				09-09-2025	

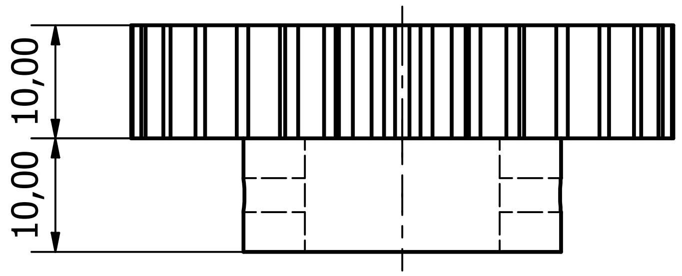
Project_Stationary_Wheel_1	Edition	Sheet 1 / 1

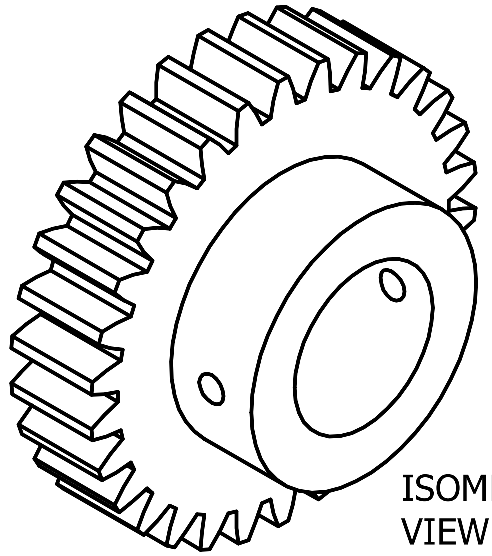
PART - 8



A-A (1.5 : 1) / .







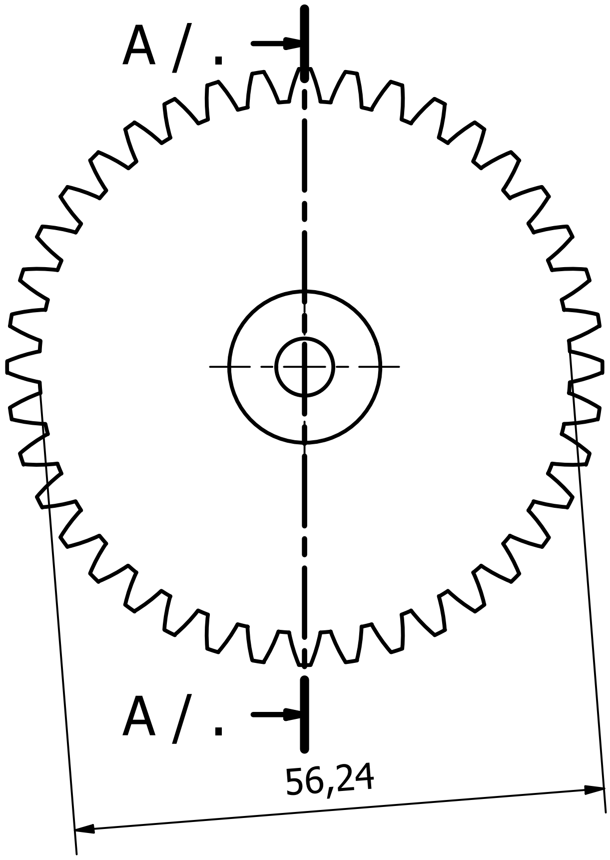
ISOMETRIC VIEW

Scale 1.5 : 1
All dimensions in mm, unless specified otherwise
Quantity: 2
Material: M.S. Rod
Process: Conventional (Drilling, Milling, Turning/Lathing, Tooth Cutting)

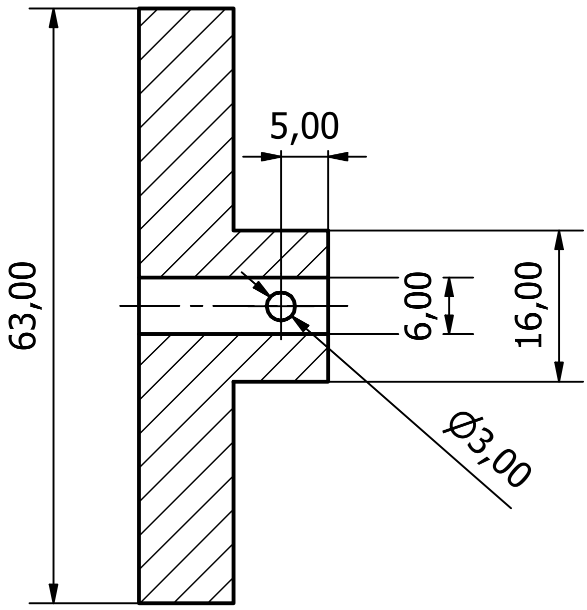
No. of Teeth (n): 30
Module (m): 1.5
Outer Diameter (OD) = m(n+2): 48
Rod Diameter (ID): 17.2
Indexing (40/n): 1 1/3

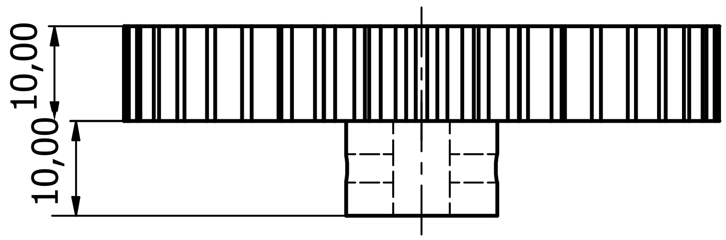
Designed by	Checked by	Approved by	Date	Date	
mukun				09-09-2025	
SpurGear30				Edition	Sheet 1 / 1

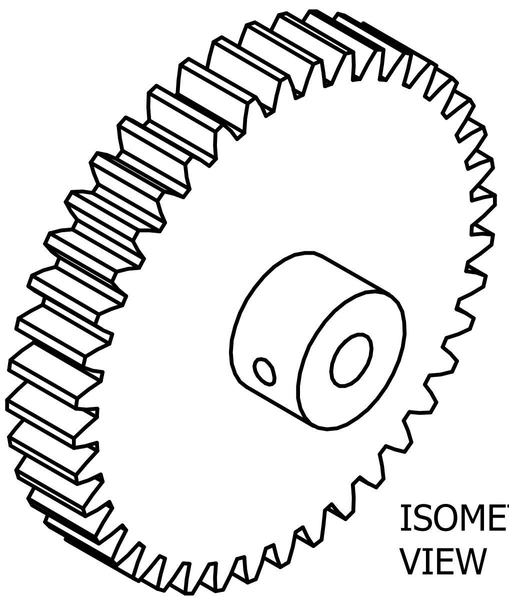
PART - 9



A-A (1.25 : 1) / .







ISOMETRIC VIEW

Scale 1.25 : 1
All dimensions in mm, unless specified otherwise
Quantity: 1
Material: M.S. Rod
Process: Conventional (Drilling, Milling, Turning/Lathing, Tooth Cutting)

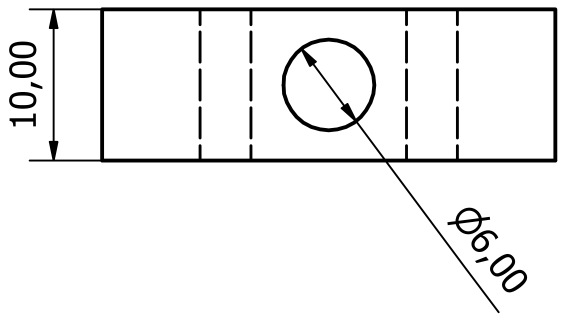
No. of Teeth (n): 40
Module (m): 1.5
Outer Diameter (OD) = m(n+2): 63
Rod Diameter (ID): 6
Indexing (40/n): 1

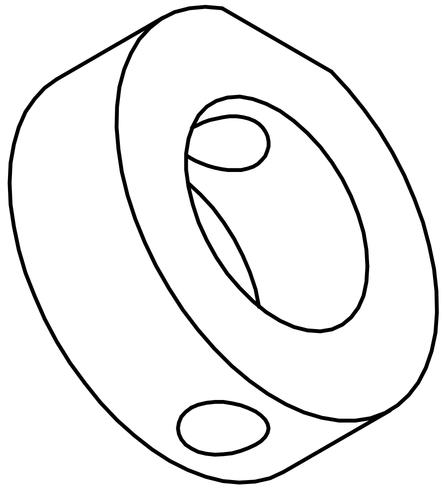
Designed by	Checked by	Approved by	Date		Date	
Mukund Advaith					09-09-2025	

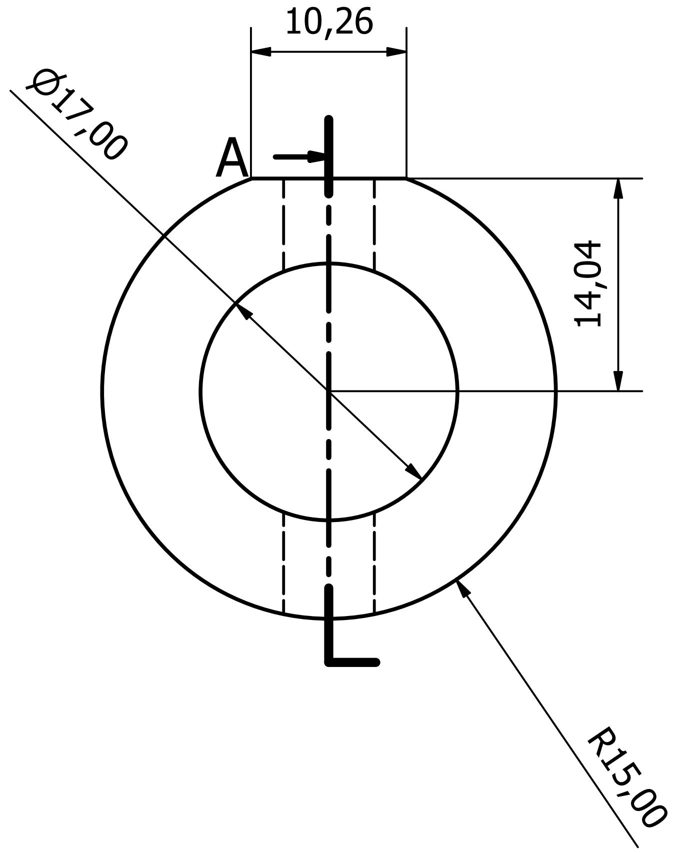
	SpurGear40	Edition	Sheet 1 / 1

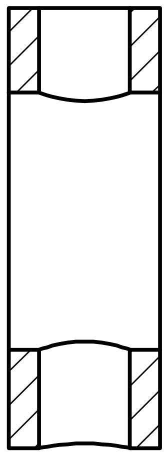
PART - 10

Dimensions in mm
SCALE: 2:1
Process: Drilling, Milling
Taper pin compatibility: 6mm









A

Material : Mild Steel
QUANTITY: 4

Designed by	Checked by	Approved by	Date		Date	
Kiran					09-09-2025	

	Edition	Sheet
Project_Stationary_rod_stopper		1 / 1